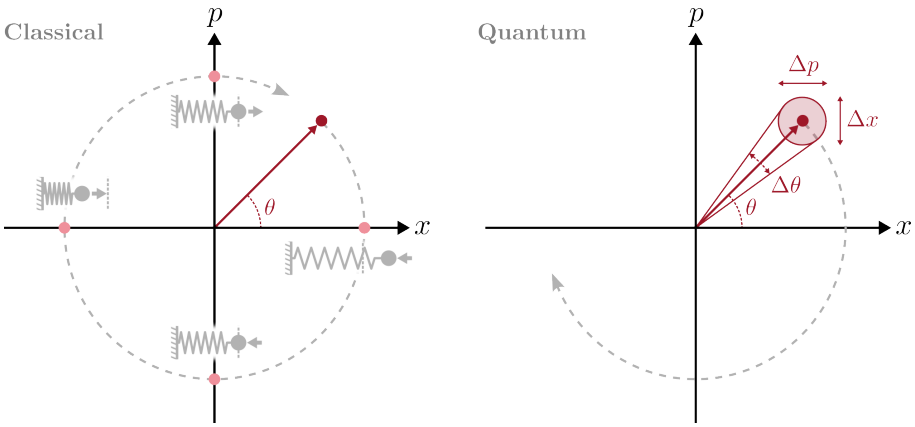

Figure 3.2: *Left:* Phase space of the classical harmonic oscillator. Both coordinates x and p are well defined at all times. In the absence of damping or driving, energy is conserved. Hence $x^2 + p^2$ is constant and the distance to the origin is fixed. The phase evolves as $\theta = -\omega t$ (see (3.76)). As the oscillator evolves through one period, the phase-space point moves around a circle. *Right:* Phase space of the quantum harmonic oscillator. There is uncertainty in the variables, as indicated by the shaded area. For a coherent state, pictured here, the uncertainty is the same along any direction and the state keeps its shape and evolves along the classical phase-space trajectory. For general states, the shape will also change with time. Note that, for the coherent state, the uncertainties are independent of the amplitude α while the distance to the phase-space origin is $\sqrt{2}|\alpha|$. The phase uncertainty $\Delta\theta$ therefore decreases with increasing mean energy, reflecting the conjugate nature of these quantities.

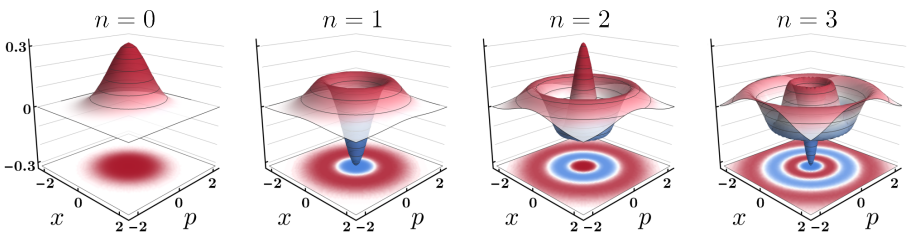

Figure 3.3: Wigner functions for the first four Fock states. Note the rotational symmetry and oscillating behaviour. The number of zero crossings equals n and the value at the origin is always $W_{|n\rangle}(0,0) = (-1)^n \pi^{-1}$.

The Wigner function

The Wigner function is perhaps the most commonly encountered quasi-probability distribution. It is widely employed in quantum optics and underlies phase-space diagrams Fig. 3.2. For a single harmonic oscillator in a state $\hat{\rho}$, the Wigner function is defined as the Wigner transform of the density operator

$$W_\rho(x,p) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \langle x + \frac{1}{2}q|\hat{\rho}|x - \frac{1}{2}q\rangle e^{ipq} dq, \tag{3.80}$$

where the ket and bra are eigenstates of $\hat{x}$. The Wigner function is real, $W_\rho(x,p)^* = W_\rho(x,p)$ and its marginals equal the probability densities for measurements of $\hat{x}$ and $\hat{p}$

$$\int_{-\infty}^{\infty} W_\rho(x,p)dp = \langle x|\hat{\rho}|x\rangle \quad \text{and} \quad \int_{-\infty}^{\infty} W_\rho(x,p)dx = \langle p|\hat{\rho}|p\rangle. \tag{3.81}$$

It follows that W_ρ is also normalised

$$\int_{-\infty}^{\infty} W_\rho(x,p)dxdp = 1. \tag{3.82}$$

When the state is pure $\hat{\rho} = |\psi\rangle\langle\psi|$ then

$$\int_{-\infty}^{\infty} W_\rho(x,p)dp = |\psi(x)|^2 \quad \text{and} \quad \int_{-\infty}^{\infty} W_\rho(x,p)dx = |\tilde{\psi}(p)|^2. \tag{3.83}$$

The Wigner function itself, however, is not generally a true probability density because it can take negative values.

The Wigner function of a Fock state $|n\rangle$ is

$$W_{|n\rangle}(x,p) = \frac{1}{\pi} e^{-x^2-p^2}(-1)^n L_n(2x^2+2p^2), \tag{3.84}$$

where $L_n(\cdot)$ is the n'th Laguerre polynomial. Plots of $W_{|n\rangle}$ for the lowest four Fock states are shown in Fig. 3.3. Note that for $n > 0$, all Fock-state Wigner functions are negative somewhere (for n odd, they are always negative at the center), and that they are rotationally symmetric around the origin, reflecting the fact that Fock states carry no phase information[4].

The Wigner function of a coherent state $|\alpha\rangle$ is the same as for the Fock state $|0\rangle$ but displaced to be centered at $(x_\alpha, p_\alpha) = (\sqrt{2}\,\mathrm{Re}(\alpha), \sqrt{2}\,\mathrm{Im}(\alpha))$. It is a Gaussian

$$W_{|\alpha\rangle}(x,p) = \frac{1}{\pi} e^{-(x-x_\alpha)^2-(p-p_\alpha)^2}, \tag{3.85}$$

with variance $\frac{1}{2}$ for both x and p. Fig. 3.4 shows plots of coherent-state Wigner functions[5]. The line indicating uncertainty in phase-space diagrams such as Fig. 3.2 can be understood as a contour line of the Wigner function of the state, e.g. the line corresponding to one standard deviation. States with Gaussian Wigner functions are known as *Gaussian states*. They are very important because they are relatively easy to generate and manipulate and have many applications and because there are powerful theoretical techniques for dealing with them in a compact manner. Gaussian states include single- and two-mode squeezed states which will be discussed later in this chapter.

For a system of N harmonic oscillators, the Wigner function generalises to

$$W_\rho(\boldsymbol{x},\boldsymbol{p}) = \frac{1}{(2\pi)^N} \int_{\boldsymbol{R}^N} \langle \boldsymbol{x} + \frac{1}{2}\boldsymbol{q}|\hat{\rho}|\boldsymbol{x} - \frac{1}{2}\boldsymbol{q}\rangle e^{i\boldsymbol{p}\cdot\boldsymbol{q}}\, d\boldsymbol{q}, \tag{3.86}$$

where $\boldsymbol{x}$ and $\boldsymbol{p}$ are N-dimensional vectors of phase-space variables, the ket of a vector should be understood as a product of $\hat{x}$-eigenstates $|\boldsymbol{z}\rangle = |z_1\rangle \cdots |z_N\rangle$, and similarly for the bra.

[4]There is no Hermitian observable corresponding to phase. Nevertheless, phase behaves roughly as a conjugate variable to the excitation number. Hence, eigenstates of $\hat{n}$ can be expected to have completely uncertain phase.

[5]Coherent states do carry phase information. The larger the amplitude of $|\alpha\rangle$, the more certain the phase (as seen in Fig. 3.2) while the uncertainty in $\hat{n}$ grows as $\langle\Delta^2\hat{n}\rangle = |\alpha|^2$.

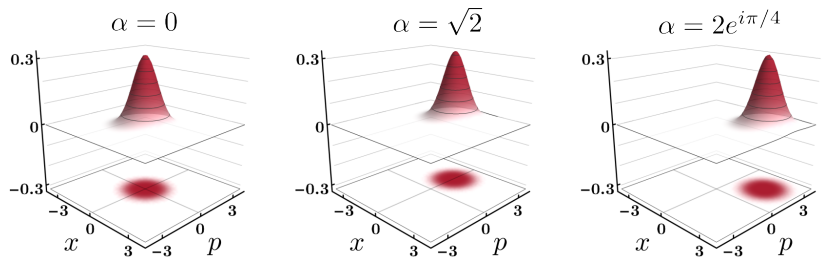

Figure 3.4: Wigner functions for coherent states with (x_α, p_α) equal to $(0,0)$, $(2,0)$, and $(2,2)$, from left to right.

The P-function

The Glauber-Sudarshan P-function (or P-representation) gives another quasiprobability distribution on phase space. Any state $\hat{\rho}$ can be expressed on a diagonal form in terms of coherent states as

$$\hat{\rho} = \int P_\rho(\alpha)|\alpha\rangle\langle\alpha|\, d^2\alpha, \tag{3.87}$$

where the integral is over the complex plane and P_ρ is the P-function. This is possible because the coherent states are not orthogonal and form an overcomplete set. P_ρ is real because $\hat{\rho}$ is Hermitian and normalised because $\int P_\rho(\alpha)\, d^2\alpha = \mathrm{Tr}(\hat{\rho}) = 1$. Like the Wigner function, however, P_ρ can become negative or highly singular for some states and thus is not a true probability distribution.

The Q-function

A third phase-space distribution, which is sometimes useful, is the Husimi Q-function, defined in terms of the expectation value of $\hat{\rho}$ with respect to a coherent state

$$Q_\rho(\alpha) = \frac{1}{\pi}\langle\alpha|\hat{\rho}|\alpha\rangle. \tag{3.88}$$

The Q-function is real, positive, and bounded $0 \leq Q_\rho(\alpha) \leq \frac{1}{\pi}$, and it is normalised because of Eq. (3.74). It is thus a proper probability distribution over the complex plane. Note that, because of (3.74), the operators

$$\hat{E}_\alpha = \frac{1}{\pi}|\alpha\rangle\langle\alpha|, \tag{3.89}$$

form a POVM (with continuous outcome). $Q_\rho(\alpha) = \mathrm{Tr}[\hat{E}_\alpha\hat{\rho}]$ is the probability for obtaining outcome α when performing the corresponding measurement on a system in state $\hat{\rho}$. The measurement is not projective because different coherent states are not orthogonal. Such a measurement can be realised in optical systems by heterodyne detection, as discussed in Sec. 3.2.5.

The characteristic function

The three quasiprobability distributions above (Wigner, P- and Q-functions) are in fact closely related and can all be expressed in terms a so-called charac-